Negative Gaussian simplex stability for sufficiently many parts

AI-assisted research manuscript

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Internally audited proof candidate for independent review. The new arguments and their assembly have undergone multiple internal reconstructions by AI agents. This is not peer review, external endorsement, or formal verification. No human authorship is asserted. The proof uses the precisely stated prior inputs in Section 2; their frozen source hashes and the new proof sources are recorded in Appendix A.

Abstract

We present a proof candidate for the single-assignment negative Gaussian simplex inequality when the number of parts is sufficiently large and the correlation belongs to the full negative voting interval. The argument reduces the remaining exterior regime to a summed radial covariance estimate. We prove that estimate using a calibrated diagonal supporting covariance, with a coordinatewise second-derivative bound that remains uniform when the total Frobenius perturbation is large. An exactly centered equal-row proxy is then used as a quadratic test, without replacing the original assignment. A Gaussian maximum comparison pays its second-chaos loss, while a dimension-free hypercontractive estimate retains the entire signed higher tail. The conclusion holds for all measurable simplex-valued assignments, with no prescribed masses, in every finite ambient dimension at least one less than the number of parts. The proof is relative to the explicitly listed previously internally audited endpoint, localization, and local comparison propositions. The comparison is direct for arbitrary competitive assignments. We also state the negative discrete plurality consequence and the exact UGC-conditional MAX-m-CUT threshold.

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1 Statement and conventions

Let γ_N be standard Gaussian measure on $\mathbb{R}^N$ and define

$$T_\rho u(x) = \mathbb{E}u(\rho x + \sqrt{1-\rho^2}\, Y), \qquad Y \sim \gamma_N, \quad -1 < \rho < 1.$$

An assignment is a measurable map $f = (f_1, \dots, f_m)$ into $\Delta_m = \{a \in [0,1]^m : \sum_i a_i = 1\}$. Its noise stability is $S_\rho(f) = \sum_i \langle f_i, T_\rho f_i \rangle_{L^2(\gamma_N)}$. This is a single-assignment functional; two independently chosen rules are not permitted in its definition.

Let $P = I_m - \mathbf{1}\mathbf{1}^T/m$. Choose vectors $b_i^0 \in \mathbb{R}^N$ whose Gram matrix is $\alpha^2 P$, where

$$n = m-1, \quad L = 1+\log m, \quad \mu_m = \mathbb{E}\max_{i\le m} Z_i, \quad \alpha = \frac{\mu_m}{m-1}, \quad w = \alpha^2(m-1), \quad \beta = \sqrt{w/m}. \tag{1.1}$$

Here the Z_i are independent standard Gaussians. The regular assignment g chooses the largest $b_i^0 \cdot x$. Ties are Gaussian null. Its centroid matrix is $B_0 = (b_i^0)_i$, and its cells have mass $1/m$. The construction exists for every finite $N \ge m-1$ and extends cylindrically when $N > m-1$.

Theorem 1.1. *There is an absolute integer m_0 such that, for every $m \ge m_0$, every finite $N \ge m-1$, every $\rho \in [-1/(m-1), 0)$, and every measurable assignment $f : \mathbb{R}^N \to \Delta_m$,*

$$S_\rho(f) \ge S_\rho(g).$$

Equality holds precisely for the indicator assignments of regular Gaussian simplex partitions, up to orthogonal transformations, relabeling, and Gaussian null sets. No mass constraint is imposed on f. The threshold m_0 is independent of ρ and of finite N.

This manuscript provides no numerical value of m_0. It does not claim the positive-correlation conjecture, the remaining small values of m, all negative Gaussian correlations, or a bilinear theorem. Section 7 gives the exact negative Plurality is Stablest consequence and the UGC-conditional Frieze–Jerrum threshold. These are fixed-m consequences, with no assertion that the ratio is minimized at the negative endpoint.

The original negative Standard Simplex formulation has no mass constraint [19, Conjecture 1.4]. The older large-m endpoint estimate in [20, Proposition 11.4], recalled in [19, equation (20)], is asymptotic, with an additive error of order $\log\log(m)/m^2$. Theorem 1.1 instead asserts exact optimality for each fixed sufficiently large m on an entire interval. The bounded primary-source check [18] is not an exhaustive priority or novelty determination.

For an assignment write

$$p_i = \mathbb{E}f_i, \qquad B_i = \mathbb{E}[X f_i], \qquad W = \|B\|_{\mathrm{F}}^2, \qquad U = m^{-1}\mathbf{1}. \tag{1.2}$$

Always $B^T\mathbf{1} = 0$. For a centered score matrix B define

$$F_p(B) = \sup\left\{ \mathbb{E}\sum_i f_i B_i \cdot X : f : \mathbb{R}^N \to \Delta_m,\ \mathbb{E}f = p \right\}, \tag{1.3}$$

$$\mathcal{R}(B) = \frac{w + \|B\|_{\mathrm{F}}^2}{2} - F_U(B). \tag{1.4}$$

Its value depends only on BB^T. Let $\Gamma(C) = F_U(C^{1/2})$ for a centered positive semidefinite matrix C; all square-root realizations are equivalent by Gaussian invariance. The regular orbit distance is $d(B) = \inf_{B_0} \|B - B_0\|_{\mathrm{F}}$, over regular centroid matrices in the same ambient space. Constants denoted c, C may change from line to line. A subscript records fixed parameter dependence. All estimates called uniform below are independent of finite N and of the correlation in the stated interval.

2 Precisely stated prior inputs

The following propositions are the previously internally audited inputs to the new proofs in Sections 3–6. They are stated with the exact hypotheses used. Their proofs are in the frozen sources listed in Appendix A; they are not replaced here by informal appeals to optimizer geometry. The new radial estimate is *not* assumed in this section.

Prior input 2.1 (Covariance value, normal strain, and offsets)**.** *The functional Γ is concave and continuous on centered positive semidefinite matrices. At a positive definite matrix on $\mathbf{1}^\perp$, its derivative is $\mathcal{L}/2$, where $\mathcal{L}$ is the mass-offset Laplacian. In particular $\Gamma(P) = \mu_m$. It is C^2 locally in the positive cone. At $C_0 = \alpha^2 P$, the normal-strain estimate implies*

$$D^2\Gamma(C_0)[E,E] < 0 \quad \text{if some } (e_i - e_j)^T E(e_i - e_j) \neq 0, \tag{2.1}$$

for a centered symmetric E. Every regular facet has positive Gaussian measure, so this strictness criterion applies to each pair.

If $\|B\|_F^2 = w$, $\max_i |B_i| \le A\alpha$, and h_U is a mean-zero balanced optimal offset, then

$$\|h_U\|^2 \le C_A L^2 \mathcal{R}(B). \tag{2.2}$$

This is a value/offset statement at arbitrary rank, including coincident sites by continuity. Positive scaling gives

$$\mathcal{R}(sB) = \tfrac{1}{2}w(s-1)^2 + s\mathcal{R}(B), \qquad h_U(sB) = sh_U(B) \quad (s > 0,\ \|B\|^2 = w). \tag{2.3}$$

In the mass box $p_i \in [1/(2m), 3/(2m)]$, the optimum first-moment value $V(p) = \sup_{\mathbb{E}f=p}\|M_f\|^2$ satisfies $V(p) \le w + CL^2\|p - U\|^2$.

These are the covariance, offset, and rank-boundary envelope results [1, 2, 3, 4].

Proposition 2.2 (Endpoint and zero-remainder rigidity)**.** *For every centered B, $\mathcal{R}(B) \ge 0$, with equality precisely when $BB^T = \alpha^2 P$. Every balanced measurable assignment has $W \le w$; equality forces the regular assignment. More generally, if an arbitrary assignment's actual centroids have $\mathcal{R}(B) = 0$, it is regular.*

Proof. We give the direct covariance proof [21]. For iid Gaussian coordinates, the expected winning coordinate is μ_m. Label symmetry gives diagonal centroid entry μ_m/m. The common coordinate mean is independent of the winning label and has mean zero. The off-diagonal entries are equal by symmetry, and centering makes the centroid matrix exactly $B_0 = \alpha P$ on $\mathbf{1}^\perp$. Its masses are $1/m$ and $F_U(B_0) = \alpha\mu_m = w$.

The reference winner is unique off Gaussian-null hyperplanes. Hence the score-value envelope is differentiable there and has derivative B_0. Differentiating $C = BB^T$ at B_0, which is invertible on $\mathbf{1}^\perp$, gives $\nabla\Gamma(C_0) = P/2$. Concavity therefore yields

$$\Gamma(C) \le w + \tfrac{1}{2}(\operatorname{tr} C - w)$$

for every centered positive semidefinite C, proving nonnegativity. If $C \neq C_0$, put $E = C - C_0$. If every edge quadratic in (2.1) vanished, writing $d_i = E_{ii}$ would give $E_{ij} = (d_i + d_j)/2$. The row sums then imply $md_i + \operatorname{tr} E = 0$ for each i, and summing forces $\operatorname{tr} E = 0$ and $E = 0$, a contradiction. Thus the path $C_0 + tE$ has strictly negative second derivative at zero, and by local C^2 regularity on a short initial interval. Its positive gap below the reference tangent is thereafter nondecreasing

by concavity. Continuity handles a singular endpoint at $t = 1$. This proves $\mathcal{R}(B) > 0$ unless $BB^T = C_0$.

For balanced f with actual centroids B, feasibility gives $W \le F_U(B) \le (w + W)/2$, hence $W \le w$. Equality implies $BB^T = C_0$. For any assignment with this actual Gram matrix, $W = w = \mathbb{E}\max_i B_i \cdot X$. Its nonnegative routing loss $\max_i B_i \cdot X - \sum_i f_i B_i \cdot X$ has expectation zero. The regular rows have Gaussian-null ties, so f is their regular winning assignment. This final implication needs no mass assumption. □

Prior input 2.3 (Competitive localization)**.** *For all sufficiently large m and $0 < r \le 1/(m-1)$, an assignment with $S_{-r}(f) \le S_{-r}(g)$ has $p_i \in [1/(2m), 3/(2m)]$. Put*

$$\delta = \|p - U\|^2, \quad \Delta = w - W, \quad D = \delta + r\Delta.$$

Then

$$0 \le \delta \le 2D, \qquad 0 \le D \le Cr^2/m. \tag{2.4}$$

This is [5], with the universal competitive-assignment scope explicitly reconstructed in [16]. For every assignment, conditional Cauchy gives $(B_i \cdot v)^2 \le p_i \mathbb{E}[f_i(X)(v \cdot X)^2]$. Summation shows

$$B^T B \preceq p_{\max} I. \tag{2.5}$$

Conditional Jensen in $\mathbb{E}e^{tv\cdot X}$ similarly gives

$$|B_i| \le p_i \sqrt{2\log(1/p_i)}. \tag{2.6}$$

Neither estimate requires stationarity or purity.

Prior input 2.4 (Global one-row covariance coercivity)**.** *For every fixed A, every centered score matrix with $\|B\|^2 = w$ and $\max_i |B_i| \le A\alpha$ satisfies*

$$\max_i \left| \frac{|B_i|^2}{\beta^2} - 1 \right|^2 \le C_A \frac{m^2}{L} \mathcal{R}(B). \tag{2.7}$$

No rank, assignment-realizability, or minimization hypothesis is imposed.

This is the global single-row estimate of [6, Theorem 1]. It will be applied directly to a normalized actual centroid matrix; no actual-minimizer row theorem is assumed.

Prior input 2.5 (Global covariance remainder and local full-noise comparison)**.** *For centered B, put $C = BB^T/\alpha^2$ and $e = \|C - P\|_{\mathrm{F}}$. The initial-segment covariance bound is*

$$\mathcal{R}(B) \ge \frac{c\alpha^2 e^2}{m} \min\left\{1, \frac{c}{Le}\right\}. \tag{2.8}$$

It holds at arbitrary endpoint rank and does not require $\|B\|^2 = w$; the value at $e = 0$ is interpreted by continuity.

For every fixed A, sufficiently large m, and every measurable assignment whose actual *centroids obey $d(B) \le A/m$ and whose masses lie in $[1/(2m), 3/(2m)]$,*

$$S_{-r}(f) - S_{-r}(g) \ge \tfrac{1}{2}\|p - U\|^2 + r\mathcal{R}(B), \qquad 0 < r \le 1/(m-1). \tag{2.9}$$

Competitiveness in this statement forces a regular assignment.

The covariance estimate follows from [7] and its unnormalized extension in [8]; the local theorem is [9]. The local theorem concerns arbitrary measurable assignments, not only affine fans.

Prior input 2.6 (Regular normalization and margin)**.** *Let Π_k denote the orthogonal Hermite projection of degree k and $A_k(f) = \sum_i \|\Pi_k f_i\|_2^2$. Then*

$$A_2(g) = \frac{m\{\mathbb{E}[(\max_i Z_i)^2] - 1\}^2}{2(m-1)(m-2)}, \qquad \alpha^2 \asymp L/m^2. \tag{2.10}$$

For a regular score matrix B_0, let $G(X)$ be its largest score minus its second largest score. Then

$$\mathbb{P}\{G \le t\} \le Cmt \quad (t \ge 0). \tag{2.11}$$

The exact second-chaos identity is in [5]; the margin and its fractional routing consequence are in [10]. We also use the standard dimension-free Gaussian hypercontractivity $\|T_t u\|_2 \le \|u\|_{1+t^2}$ [26], and the one-dimensional variance bound $\mathrm{Var}_\nu\, x \le 1/c$ when the density of ν has logarithmic second derivative at most $-c$ [24]. The transitive proofs of the offset and mass-window inputs also use Gaussian isoperimetry [25] and the convex Ehrhard inequality [23]; their precise uses are recorded in the companion section-level source map.

Prior input 2.7 (Equal-threshold Gaussian maximum comparison)**.** *If K is an $m \times m$ correlation matrix and $K - \mathbf{1}\mathbf{1}^T/m \succeq 0$, then for $Y \sim N(0, K)$ and independent standard Gaussians Z_i,*

$$\mathbb{P}\{\max_i Y_i \le t\} \ge \Phi(t)^m \qquad (t \in \mathbb{R}).$$

The statement includes singular K.

This is Theorem 2.1 of the pinned primary source [15]. Only equal thresholds are used. We do not infer a general rectangular comparison or a comparison for unequal marginal variances.

3 Calibrated diagonal derivatives on a coordinate box

This section proves the new derivative estimate of [11]. The distinction between a small maximum coordinate and a small Frobenius norm is essential.

Proposition 3.1. *Fix $A > 0$. Let $\sum_i \xi_i = 0$ and $\max_i |\xi_i| \le A$. Set $v_i(t) = 1 + t\xi_i$, $\sigma_i = \sqrt{v_i}$, and choose the unique mean-zero offsets $h(t)$ such that the independent scores $\sigma_i Z_i + h_i$ give every label probability $1/m$. For all sufficiently large m, on $0 \le t \le c_A/L^2$, define*

$$\begin{aligned} F_i(x) &= \Phi((x - h_i)/\sigma_i), & f_i(x) &= \partial_x F_i(x), & a_i(x) &= f_i(x)/F_i(x), \\ W_0(x) &= \prod_i F_i(x), & A_h(x) &= \sum_i a_i(x), & \rho_{ij}(x) &= W_0(x) a_i(x) a_j(x), \\ \omega_{ij} &= \int_{\mathbb{R}} \rho_{ij}(x)\,\mathrm{d}x, & \ell_i(t) &= \sum_j \omega_{ij} = \int_{\mathbb{R}} W_0 A_h a_i\,\mathrm{d}x. \end{aligned}$$

The analytic quantity ω_{ii} is included in ℓ_i but is not a physical facet weight. Put $M_1 = m^{-1}\sum_i |\xi_i|$ and $M_2 = m^{-1}\sum_i \xi_i^2$. Then

$$c\alpha/m \le \omega_{ij} \le C\alpha/m, \tag{3.1}$$

$$|h_i'| \le C\sqrt{L}(|\xi_i| + M_1), \qquad |h_i''| \le CL^{3/2}(\xi_i^2 + M_2), \tag{3.2}$$

$$|\ell_i''(t)| \le C\alpha L^2(\xi_i^2 + M_2). \tag{3.3}$$

The constants depend only on A, not on $\|\xi\|_2$. If T has density proportional to $\phi(x)^2\Phi(x)^{m-2}$, then

$$\ell_i(0) = \alpha, \qquad \ell_i'(0) = -\alpha\eta_m\xi_i, \qquad \eta_m = \frac{m-1-m\operatorname{Var} T}{2(m-2)} \in [1/4, 1/2]. \tag{3.4}$$

In particular,

$$|\ell_i(t) - \alpha + \alpha\eta_m t\xi_i| \le \tfrac{1}{2}C\alpha L^2t^2(\xi_i^2 + M_2). \tag{3.5}$$

3.1 Uniform threshold moments

We first prove the analytic estimate needed for all differentiations.

Lemma 3.2. *For a fixed small $\delta_0 > 0$, suppose*

$$\epsilon := \max_i |v_i - 1| \le \delta_0/L, \qquad H := \max_i |h_i| \le \delta_0/\sqrt{L}. \tag{3.6}$$

Balance is not required here. For every fixed pair of nonnegative integers p, q, uniformly in all i, j, including $i = j$,

$$\omega_{ij} \asymp \alpha/m, \qquad \mathbb{E}_{ij}\Big[(m\max_k a_k)^p(1+|x|)^q\Big] \le C_{p,q}L^{(p+q)/2}, \tag{3.7}$$

where $\mathbb{E}_{ij}$ is expectation under ρ_{ij}/ω_{ij}.

Proof. Write

$$Z_0 = \int \phi(x)^2\Phi(x)^{m-2}\,\mathrm{d}x = \frac{\mu_m}{m(m-1)} = \frac{\alpha}{m} \asymp \frac{\sqrt{L}}{m^2}. \tag{3.8}$$

The identity follows by integration by parts in $\mu_m = m\int x\phi(x)\Phi(x)^{m-1}\,\mathrm{d}x$. The order $\mu_m \asymp \sqrt{L}$ follows from the Gaussian union bound for the upper estimate and the usual Mills lower bound at a fixed fraction of $\sqrt{2\log m}$ for the lower estimate.

For $x \ge 2H$, set $z = (x+H)/\sigma_{\min} \ge 0$. Every standardized argument $y_k = (x-h_k)/\sigma_k$ is nonnegative and belongs to $[z - C\epsilon z - CH, z]$. Comparing Gaussian densities gives

$$\rho_{ij}(x)\,\mathrm{d}x \le C\phi(z)^2\Phi(z)^{m-2}e^{C\epsilon z^2+CHz+CH^2}\,\mathrm{d}z, \tag{3.9}$$

$$\max_k a_k(x) \le C\frac{\phi(z)}{\Phi(z)}e^{C\epsilon z^2+CHz+CH^2}. \tag{3.10}$$

For $i = j$ the extra factor F_i^{-1} is at most 2, so the same estimates apply. For a lower weight bound instead use $y = (x-H)/\sigma_{\max}$ on $x \ge 2H$. The lower density is a constant times $\phi(y)^2\Phi(y)^{m-2}e^{-C\epsilon y^2-CHy-CH^2}$. For $i = j$, use $\Phi(y)^{m-1} \ge \frac{1}{2}\Phi(y)^{m-2}$. The omitted interval near zero has exponentially small baseline mass. The baseline law has $\mathbb{E}T^2 \le CL$: below a fixed multiple of $\sqrt{L}$ this is immediate, and above it the factor ϕ^2, divided by (3.8), gives an integrable Gaussian tail; the negative half-line is exponentially suppressed by Φ^{m-2}. Jensen's inequality therefore bounds the lower integral below by cZ_0.

For the upper moments, put $s = 1-\Phi(z)$ on $z \ge 0$. Mills bounds give $\phi(z) \le Cs\sqrt{\log(e/s)}$ and $z^2 \le 2\log(1/s)$. Also $CHz \le CH\sqrt{L}/2 + CHz^2/(2\sqrt{L})$. Thus (3.9)–(3.10), including the absolute Jacobian $|\,\mathrm{d}z| = |\,\mathrm{d}s|/\phi(z)$, bound the desired unnormalized moment by

$$C_{p,q}m^p\int_0^{1/2}(1-s)^{m-p-2}s^{p+1-2\theta}[\log(e/s)]^{(p+q+1)/2}\,\mathrm{d}s, \quad 0 \le \theta \le C_{p,q}\delta_0/L. \tag{3.11}$$

For large m, substitution $u = ms$ and splitting at $u = 1$ give

$$\int_0^{1/2} (1-s)^{m-p-2} s^{p+1-2\theta} [\log(e/s)]^b \, \mathrm{d}s \le C_{p,q} m^{-p-2+2\theta} L^b.$$

On $u \le 1$ use integrability of a fixed positive power times $\log(1/u)^b$; on $u \ge 1$ use $(1-s)^{m-p-2} \le e^{-ms/2}$. Since $m^{2\theta}$ is bounded, division by Z_0 proves (3.7) on $x \ge 2H$.

On $x \le -2H$, the product W_0 is at most $2^{-m} e^{-cmx^2}$ and every hazard is at most $C(1+|x|)$. On $[-2H, 2H]$, every F_i is bounded by a fixed number below one, and the hazards are bounded. These two regions contribute at most $C_{p,q} m^p c_0^m$ for some $c_0 < 1$, negligible relative to Z_0. This also verifies all negative-threshold tails and completes the proof. □

3.2 A componentwise inverse and the calibrated acceleration

The balanced offset exists uniquely in the mean-zero gauge: the Gaussian offset dual is coercive there, its gradient is the mass vector minus U, and its Hessian is the complete Laplacian with positive weights ω_{ij}. These integral weights are smooth for positive variances, so the implicit function theorem gives smooth $h(t)$.

Lemma 3.3. *If $c\alpha/m \le \omega_{ij} \le C\alpha/m$ for $i \ne j$, the associated Laplacian $\mathcal{L}$ satisfies, for mean-zero b,*

$$|(\mathcal{L}^{-1} b)_i| \le \frac{C}{\alpha} \left(|b_i| + \frac{1}{m} \sum_j |b_j| \right). \tag{3.12}$$

Proof. Subtract $c\alpha/m$ from every edge weight. The remainder is a graph Laplacian $\mathcal{L}_0$, and on $\mathbf{1}^\perp$, $e^{-s\mathcal{L}} = e^{-c\alpha s} e^{-s\mathcal{L}_0}$. The second semigroup is doubly stochastic, hence contracts ℓ^1. Integration gives $\|\mathcal{L}^{-1} b\|_1 \le \|b\|_1/(c\alpha)$. For $y = \mathcal{L}^{-1} b$, the row equation $\deg_i y_i = b_i + \sum_{j \ne i} \omega_{ij} y_j$, together with $\deg_i \ge c\alpha$ and the upper weight bound, proves (3.12). □

Define the threshold velocities and logarithmic density derivatives

$$b_i(x) = h_i' + \frac{\xi_i}{2v_i}(x - h_i), \tag{3.13}$$

$$L_i(x) = \partial_t \log f_i = -\frac{\xi_i}{2v_i} + \frac{\xi_i (x-h_i)^2}{2v_i^2} + \frac{h_i'(x-h_i)}{v_i}. \tag{3.14}$$

The density transport equation $\partial_t f_i = -\partial_x (f_i b_i)$ and integration by parts in the winning-probability integral give

$$0 = p_i' = \sum_{j \ne i} \int \rho_{ij} (b_i - b_j) \, \mathrm{d}x. \tag{3.15}$$

While (3.6) holds, the threshold moment lemma and Lemma 3.3 solve this equation for h' and give the first estimate in (3.2).

This estimate closes the box by continuation. Starting with $h(0) = 0$, it gives $\|h(t)\|_\infty \le C_A t\sqrt{L}$. Choose a bootstrap bound twice this size and then choose c_A small. For $t \le c_A/L^2$ the offset and variance bounds lie strictly inside (3.6). The constants in that box do not depend on the bootstrap constant. Thus the first derivative estimate and the weight comparison hold on the whole interval, without smallness of $\|h\|_2$ or $\|\xi\|_2$.

For the second derivative set

$$S(x) = \sum_k a_k b_k, \qquad P_{ij} = L_i + L_j + a_i b_i + a_j b_j - S, \tag{3.16}$$

$$c_i(x) = -\frac{\xi_i^2 (x - h_i)}{2v_i^2} - \frac{\xi_i h_i'}{2v_i}. \tag{3.17}$$

Then $\partial_t \log \rho_{ij} = P_{ij}$, also for $i = j$, and $b_i' = h_i'' + c_i$. For every fixed finite k, the moment lemma gives

$$\|b_i\|_{L^k(ij)} \le C_k \sqrt{L}(|\xi_i| + M_1), \tag{3.18}$$

$$\|P_{ij}\|_{L^k(ij)} \le C_k L(|\xi_i| + |\xi_j| + M_1). \tag{3.19}$$

For the summed term use $\sum_k a_k(|\xi_k| + M_1) \le 2m \max_k a_k M_1$. This step retains the contributions of all other labels without losing a factor m.

Differentiating (3.15) yields exactly

$$\mathcal{L} h'' = -b^{(2)}, \qquad b_i^{(2)} = \sum_{j \ne i} \int \rho_{ij} \{P_{ij}(b_i - b_j) + c_i - c_j\} \, \mathrm{d}x. \tag{3.20}$$

Hölder's inequality, (3.19), and $M_1^2 \le M_2$ give $|b_i^{(2)}| \le C\alpha L^{3/2}(\xi_i^2 + M_2)$ and $m^{-1} \sum_i |b_i^{(2)}| \le C\alpha L^{3/2} M_2$. The right side has mean zero by pairwise antisymmetry. Applying Lemma 3.3 proves the asserted bound for h_i''.

3.3 Second derivatives of the hazard coefficients

Since $\partial_t^2 \rho_{ij} = \rho_{ij}(P_{ij}^2 + P_{ij}')$, it remains to estimate the two terms. Differentiating (3.14) gives

$$L_i' = \frac{\xi_i^2}{2v_i^2} - \frac{\xi_i^2 (x - h_i)^2}{v_i^3} - \frac{2\xi_i h_i' (x - h_i)}{v_i^2} + \frac{h_i''(x - h_i)}{v_i} - \frac{(h_i')^2}{v_i}.$$

Its conditional L^k norm is at most $C_k L^2(\xi_i^2 + M_2)$. Moreover

$$a_i' = a_i(L_i + a_i b_i), \qquad (a_i b_i)' = a_i\{(L_i + a_i b_i) b_i + b_i'\}.$$

The same bound holds for this last expression. For S' first sum pointwise using

$$\sum_k a_k \xi_k^2 \le m \max_k a_k M_2, \qquad \sum_k a_k (|\xi_k| + M_1)^2 \le Cm \max_k a_k M_2.$$

An additional factor $\max a_k$ handles the terms with a_k^2. Lemma 3.2 then yields $\|S'\|_{L^k(ij)} \le C_k L^2 M_2$. Consequently

$$\mathbb{E}_{ij}(P_{ij}^2 + |P_{ij}'|) \le CL^2(\xi_i^2 + \xi_j^2 + M_2). \tag{3.21}$$

All derivatives are dominated by the integrable functions just estimated, so differentiation under the integral is legitimate. Summing (3.21) with $\omega_{ij} \le C\alpha/m$, including $j = i$, proves (3.3).

3.4 The exact regular coefficient

At $t = 0$, all variances are one and all offsets are zero. Equation (3.15) gives $h_i'(0) = -(\mathbb{E}T)\xi_i/2$. Put $M = \mathbb{E}T$ and $a(x) = \phi(x)/\Phi(x)$. Then

$$\partial_t \log a_i|_{t=0} = \frac{(x + a(x))(x - M) - 1}{2}\,\xi_i =: k(x)\xi_i.$$

Because $\sum_i \xi_i = 0$, both $\partial_t W_0$ and $\partial_t A_h$ vanish at zero. Thus $\ell_i'(0) = \alpha \mathbb{E}_T k(T)\xi_i$, and $\ell_i(0) = mZ_0 = \alpha$.

The logarithmic derivative of the tie density is $-2x + (m-2)a(x)$. Integration by parts against $x - M$ gives

$$-2\operatorname{Var} T + (m-2)\operatorname{Cov}(T, a(T)) = -1.$$

It follows that $\mathbb{E}k(T) = [m \operatorname{Var} T - m + 1]/[2(m-2)]$. The tie density has logarithmic second derivative at most -2, so $\operatorname{Var} T \le 1/2$. For the reverse bound, decompose the independent regular scores into their common mean $\overline{Z}$ and their centered vector. Conditioning on $Z_i = Z_j$ and on their being maximal depends only on the centered vector. The raw threshold is $T = \overline{Z} + T_H$, with independent $\operatorname{Var}\overline{Z} = 1/m$. Hence $\operatorname{Var} T \ge 1/m$. This proves (3.4) and its coefficient interval. Integrating (3.3) twice proves (3.5), completing Proposition 3.1.

Remark 3.4 (Normalization check). The physical degree is $\ell_i - \omega_{ii}$. At zero, $\omega_{ii}' = -2\alpha\eta_m\xi_i/m$. Therefore Price's formula gives, in variance coordinates,

$$F_v''[\xi, \xi] = -\frac{\alpha}{4}\left(\frac{m-1}{m} - \operatorname{Var} T\right)\|\xi\|_2^2.$$

In standard-deviation coordinates the corresponding balanced Hessian is $\alpha \operatorname{Var} T\|\xi\|_2^2$. These local checks do not replace the uniform coordinatewise remainder proved above.

4 Global summed radial coercivity

We now prove the supporting-covariance theorem of [12]. Only the supporting point is near the regular covariance; the target may be singular and far from it.

Theorem 4.1. *Fix $A \ge 1$. For sufficiently large m, every centered score matrix B with $\|B\|_{\mathrm{F}}^2 = w$ and $|B_i|^2 \le A\alpha^2$ satisfies*

$$\mathcal{R}(B) \ge \frac{c_A \alpha^2}{L^2} \sum_i \left(\frac{|B_i|^2}{\alpha^2} - \frac{n}{m}\right)^2, \tag{4.1}$$

$$\sum_i (|B_i|^2 - w/m)^2 \le C_A \frac{L^3}{m^2}\mathcal{R}(B). \tag{4.2}$$

No lower row bound, pair separation, rank, or proximity assumption is required.

Proof. For positive v_i let $D = P\operatorname{diag}(v_i)P$. Independent scores $\sqrt{v_i}Z_i$, after subtracting their common sample mean, realize D. This subtraction changes neither the allocation nor its expected value. Using balanced offsets and the notation of Proposition 3.1, the mass-offset Laplacian is

$$\mathcal{L}_D = P\operatorname{diag}(\ell_i)P - H_D, \qquad H_D = \int W_0(Pa)(Pa)^T \,\mathrm{d}x \succeq 0. \tag{4.3}$$

Indeed $\operatorname{diag}(\ell) - \int W_0 a a^T$ is exactly the ordinary Laplacian, which already annihilates $\mathbf{1}$.

At this positive supporting point, Prior input 2.1 gives $\nabla\Gamma(D) = \mathcal{L}_D/2$. Homogeneity of degree $1/2$ gives $\Gamma(D) = \operatorname{tr}(\mathcal{L}_D D)$. Concavity therefore implies, for every centered PSD target C,

$$\begin{aligned}\Gamma(C) &\le \Gamma(D) + \tfrac{1}{2}\operatorname{tr}[\mathcal{L}_D(C-D)] = \tfrac{1}{2}\operatorname{tr}[\mathcal{L}_D(C+D)] \\ &\le \tfrac{1}{2}\sum_i \ell_i(C_{ii}+D_{ii}).\end{aligned} \tag{4.4}$$

The discarded term is $-\operatorname{tr}[H_D(C+D)]/2 \le 0$. This is an exact PSD sign, not an entrywise comparison. Continuity gives the same support inequality for singular C.

Set $C = BB^T/\alpha^2$, so $\operatorname{tr} C = n$ and $C_{ii} \le A$. Write

$$z_i = C_{ii} - n/m, \qquad \kappa = 1 - 2/m, \qquad \xi_i = z_i/\kappa.$$

Then $\sum_i \xi_i = 0$, $\|\xi\|_\infty \le A_1 := 3(A+1)$, and

$$D_t = P + tP\operatorname{diag}(\xi)P, \qquad \operatorname{tr} D_t = n, \qquad (D_t)_{ii} = n/m + \kappa t\xi_i = n/m + tz_i. \tag{4.5}$$

Let c_H, C_H be the constants from Proposition 3.1 for A_1. Choose

$$0 < b_A \le \min\left\{c_H, 1, \frac{1}{2A_1}, \frac{1}{24C_H(A+2)}\right\}, \qquad t = b_A/L^2.$$

The supporting covariance is positive definite on $\mathbf{1}^\perp$, its diagonal is at most 2, and the whole segment lies in the proved coordinate box. Taylor's formula gives

$$\ell_i(t) = \alpha - \alpha\eta_m t\xi_i + e_i, \qquad |e_i| \le \tfrac{1}{2}C_H\alpha L^2t^2(\xi_i^2 + m^{-1}\|\xi\|^2).$$

Inserting this in (4.4), the constant term is $\alpha n = \mu_m$. The linear term pairs all diagonal errors at once:

$$\sum_i \xi_i(C_{ii} + (D_t)_{ii}) = \kappa(1+t)\|\xi\|^2.$$

The absolute remainder is at most $C_H(A+2)\alpha L^2t^2\|\xi\|^2/2$. Since $\eta_m \ge 1/4$ and $\kappa \ge 1/3$,

$$\begin{aligned}\mu_m - \Gamma(C) &\ge \alpha\left\{\frac{\eta_m\kappa t(1+t)}{2} - \frac{C_H(A+2)L^2t^2}{2}\right\}\|\xi\|^2 \\ &\ge \frac{\alpha\kappa t}{16}\|\xi\|^2 \ge \frac{\alpha b_A}{16L^2}\|z\|^2.\end{aligned}$$

For the middle inequality, after factoring αt, the error is at most $1/48$, and $\kappa/8 - 1/48 \ge \kappa/16$. Finally $\mathcal{R}(B) = w - \alpha\Gamma(C) = \alpha[\mu_m - \Gamma(C)]$. This proves (4.1); multiplying by α^4 and using $\alpha^2 \asymp L/m^2$ gives (4.2). □

Corollary 4.2 (Bounded unnormalized scales)**.** *Fix $0 < s_0 \le s_1 < \infty$ and A. If B is centered, $s = \|B\|/\sqrt{w} \in [s_0, s_1]$, and $|(B/s)_i|^2 \le A\alpha^2$, then*

$$\sum_i (|B_i|^2 - W/m)^2 + \sum_i (|B_i|^2 - w/m)^2 \le C\frac{L^3}{m^2}\mathcal{R}(B), \tag{4.6}$$

$$E_{\mathrm{rad}} := \sum_i (|B_i| - \beta)^2 \le CL^2\mathcal{R}(B). \tag{4.7}$$

Proof. Put $\widehat{B} = B/s$. Equation (2.3) is exact. The vector $(|B_i|^2 - W/m)_i$ equals s^2 times its hatted version, so Theorem 4.1 bounds its squared norm by $Cs^3\alpha^2L^2\mathcal{R}(B)$. Orthogonal decomposition into the mean and mean-zero parts gives

$$\sum_i (|B_i|^2 - w/m)^2 = \sum_i (|B_i|^2 - W/m)^2 + \frac{(W-w)^2}{m}.$$

The last term is $w^2(s+1)^2(s-1)^2/m \le 2w(s+1)^2\mathcal{R}(B)/m$. This proves (4.6). Finally $(|B_i|-\beta)^2 \le (|B_i|^2 - \beta^2)^2/\beta^2$ and $\beta^2 \asymp L/m^2$ give (4.7). □

5 An exactly centered equal-row test matrix

The matrix constructed next is only a test object. We will neither route the assignment by it nor claim it is a matrix of actual centroids.

Lemma 5.1. *Suppose $\widehat{B}$ is centered, $\|\widehat{B}\|^2 = w$, $\|\widehat{B}\|_{\mathrm{op}}^2 \le C/m$, and $\max_i||\widehat{B}_i|/\beta - 1| = o(1)$. Let $E = \sum_i(|\widehat{B}_i| - \beta)^2$ and suppose $E \le CrL^2/m$ for $0 < r \le 1/(m-1)$. For sufficiently large m there is a matrix Z, in the same row span, such that*

$$\sum_i Z_i = 0, \qquad |Z_i| = \beta, \qquad \|Z - \widehat{B}\|^2 \le CE. \tag{5.1}$$

Proof. Write $u_i = \widehat{B}_i/|\widehat{B}_i|$. The row lower bound and operator estimate give $\|m^{-1}\sum_i u_i u_i^T\|_{\mathrm{op}} \le C/L$. On the row span consider $\Psi(c) = m^{-1}\sum_i|\widehat{B}_i - c|$. For $|c| \le c_0\beta$, distances are comparable to β, and

$$\left\| \frac{1}{m}\sum_i \frac{(\widehat{B}_i - c)(\widehat{B}_i - c)^T}{|\widehat{B}_i - c|^2} \right\|_{\mathrm{op}} \le C/L + C|c|^2/\beta^2.$$

Choose a fixed sufficiently small c_0. Then $\nabla^2\Psi \succeq c\beta^{-1}I$ on that ball. At zero, centering gives

$$g_0 := \|\nabla\Psi(0)\| \le \frac{C\|\widehat{B}\|_{\mathrm{op}}\sqrt{E}}{m\beta^2} \le C\frac{\sqrt{m}\sqrt{E}}{L} \le C\sqrt{r} = o(1).$$

The radial derivative is positive on the ball boundary for large m; therefore Ψ has an interior minimizer c, with $|c| \le C\beta g_0$ and $\sum_i(\widehat{B}_i - c)/|\widehat{B}_i - c| = 0$. Define $Z_i = \beta(\widehat{B}_i - c)/|\widehat{B}_i - c|$. This gives the first two assertions in (5.1). Moreover

$$m|c|^2 \le CE/L.$$

Compare Z_i first with βu_i. Radial projection is uniformly Lipschitz on these points away from zero, so the squared comparison cost is $Cm|c|^2$; the squared cost from βu_i to $\widehat{B}_i$ is exactly E. This proves the final assertion. All operations take place in the row span. □

Remark 5.2. The proof of Lemma 5.1 is the geometric-median construction from [6], with its small-gradient condition supplied here by the new radial bound. Neither duplicate rows of Z nor rank loss will cause a problem in the quadratic test below.

6 The exterior comparison and the full signed objective

We prove the exterior calculation of [13], with its radial premise supplied by Theorem 4.1, using the direct competitive argument [16]. Let f be any measurable assignment satisfying $S_{-r}(f) \le S_{-r}(g)$, and put $\rho = -r$. Use the notation of Prior input 2.3, and write $R = \mathcal{R}(B)$ and $d = d(B)$. If $d \le 1/m$, (2.9) proves the desired conclusion. Throughout this section assume

$$d > 1/m. \tag{6.1}$$

6.1 The normalized cost and its exterior floor

From $0 \le \delta \le 2D$, one has $r|\Delta| = |D - \delta| \le D$; therefore $W/w = 1 + O(r/L)$, including when $W > w$. Since f is admissible for $F_p(B)$, $F_p(B) \ge W$. The balanced optimal offsets are a dual test at p, so

$$R \le \frac{w - W}{2} - h_U \cdot (p - U) \le \frac{\Delta}{2} + CL\sqrt{R\delta}.$$

Here (2.2) is scaled using (2.3), and the entropy bound gives the required fixed row cap. Young's inequality implies, for large m,

$$0 \le R \le \Delta + CL^2\delta \le \Delta + \delta/r = D/r =: J. \tag{6.2}$$

The division by r is indispensable.

Let $q = d/\alpha$ and $e = \|BB^T/\alpha^2 - P\|_{\mathrm{F}}$. If s_i are the singular values of B/α on $\mathbf{1}^\perp$, including zeros, Procrustes minimization gives $q^2 = \sum_i (s_i - 1)^2$ and $e^2 = \sum_i (s_i^2 - 1)^2$; hence $e \ge q$. Also $c/\sqrt{L} \le q \le C\sqrt{m}$ by (6.1) and $W \asymp w$. Equation (2.8) gives

$$J \ge R \ge \frac{cq}{m^3} \ge \frac{c}{m^3\sqrt{L}}, \qquad d^2 \le Cm^{3/2}LR. \tag{6.3}$$

These estimates concern the score remainder, not a replacement of the actual assignment by an affine one.

Put $s = \sqrt{W/w}$ and $\widehat{B} = B/s$. By Corollary 4.2, $E_{\mathrm{rad}}(\widehat{B}) \le CL^2R \le CrL^2/m \le CL^2/m^2$. The radial identity gives $\mathcal{R}(\widehat{B}) \le CR$, and (2.7) yields

$$\max_i \left| \frac{|\widehat{B}_i|^2}{\beta^2} - 1 \right| \le C\sqrt{\frac{m^2R}{L}} \le C\sqrt{\frac{mr}{L}} = o(1).$$

The universal operator bound (2.5) scales to $\|\widehat{B}\|_{\mathrm{op}}^2 \le C/m$. Thus every hypothesis of Lemma 5.1 is proved for this arbitrary competitive assignment. It gives an exactly centered equal-row matrix Z, with median gradient at most $C\sqrt{r}$. Since $\|B - \widehat{B}\|^2 = w(s-1)^2 \le 2R$,

$$E_Z := \|Z - B\|^2 \le CL^2R \le CL^2J. \tag{6.4}$$

6.2 A maximum comparison with an explicit absolute tail

Write $Y_i = Z_i \cdot X/\beta$, $M = \max_i Y_i$, and let M_* be the regular unit-row maximum. Centering gives $M \ge 0$ pointwise. Set $a_0 = (m-1)/m$ and let $\nu = \sqrt{a_0}M + G/\sqrt{m}$, with G independent standard Gaussian. The underlying Gaussian score covariance is $a_0C + \mathbf{1}\mathbf{1}^T/m$, where $C = ZZ^T/\beta^2$. It has unit diagonal and dominates $\mathbf{1}\mathbf{1}^T/m$. The regular version has covariance I_m. Prior input 2.7 therefore gives

$$H(t) := \mathbb{P}\{\nu \le t\} - \mathbb{P}\{\max_i Z_i^{\mathrm{iid}} \le t\} \ge 0.$$

Set $\varepsilon_{\text{free}} = \mathbb{E}M_* - \mathbb{E}M \geq 0$. Integration of the CDFs and cancellation of the independent common Gaussian yield

$$\int H = \sqrt{a_0}\varepsilon_{\text{free}}, \qquad \mathbb{E}M_*^2 - \mathbb{E}M^2 = \frac{2}{a_0}\int tH(t)\,\mathrm{d}t.$$

For $t \geq T$, $H(t) \leq m\overline{\Phi}(t)$; for $t \leq -T$, $H(t) \leq \Phi(t)$. Splitting the integral and using $H \geq 0$ gives, for $T \geq 1$,

$$(\mathbb{E}M_*^2 - \mathbb{E}M^2)_+ \leq CT\varepsilon_{\text{free}} + Cme^{-T^2/2}. \tag{6.5}$$

No deficit-relative tail estimate is assumed. With $V = \beta M = \max_i Z_i \cdot X$ and $V_* = \beta M_*$, $\mathbb{E}V_* = w$ and (6.5) becomes

$$(\mathbb{E}V_*^2 - \mathbb{E}V^2)_+ \leq C\beta T(w - \mathbb{E}V) + C\beta^2 me^{-T^2/2}. \tag{6.6}$$

In particular $\mathbb{E}V \leq w$.

6.3 The actual assignment pays its test-score error

Write $z_i(X) = Z_i \cdot X$ and $c_f(X) = V - \sum_i f_i z_i \geq 0$. Both $\mathbb{E}c_f$ and $w - \mathbb{E}V$ are nonnegative, and their sum is exactly

$$w - \langle Z, B\rangle = \frac{\Delta + E_Z}{2} \leq CL^2 J. \tag{6.7}$$

Here and below $\langle Z, B\rangle = \sum_i Z_i \cdot B_i$. This identity is valid even if $\Delta < 0$ and the masses are unequal. Pointwise, $z_i \leq V$ and $V \geq 0$, so

$$V^2 - \sum_i f_i z_i^2 = \sum_i f_i(V - z_i)(V + z_i) \leq 2Vc_f.$$

If $A_X = \max_i |Y_i|$, then $c_f \leq 2\beta A_X$ and $V \leq \beta A_X$. Unit marginal Gaussian tails imply

$$\mathbb{E}[A_X^2 \mathbf{1}_{\{A_X > T\}}] \leq Cm(T^2 + 1)e^{-T^2/2}.$$

Splitting according to $V \leq \beta T$ gives

$$\mathbb{E}V^2 - \mathbb{E}\sum_i f_i z_i^2 \leq 2\beta T\mathbb{E}c_f + C\beta^2 m(T^2 + 1)e^{-T^2/2}. \tag{6.8}$$

Take $T = \sqrt{32L}$. Combining (6.6)–(6.8) and $\beta T \leq CL/m$ gives

$$\mathbb{E}\sum_i f_i z_i^2 \geq \mathbb{E}V_*^2 - \frac{CL^3 J}{m} - C\beta^2 mLe^{-16L}. \tag{6.9}$$

6.4 A quadratic certificate with unequal actual masses

Define

$$Q_i = \mathbb{E}[f_i(XX^T - I)], \qquad T_i = Z_i Z_i^T - \frac{Z^T Z}{m}.$$

Since $\sum_i Q_i = 0$, every test row has squared norm β^2, and $\sum_i p_i = 1$, the numerator is

$$T_f := \sum_i \langle Q_i, T_i\rangle_{\mathrm{F}} = \mathbb{E}\sum_i f_i z_i^2 - \sum_i p_i |Z_i|^2 = \mathbb{E}\sum_i f_i z_i^2 - \beta^2. \tag{6.10}$$

Thus exact balance is not required. The denominator satisfies

$$\sum_i \|T_i\|_{\mathrm{F}}^2 = m\beta^4 - \frac{\mathrm{tr}((ZZ^T)^2)}{m} \le D_* := \frac{\alpha^4(m-1)(m-2)}{m}, \tag{6.11}$$

because $\mathrm{tr}(ZZ^T) = w$ and $\mathrm{rank}(ZZ^T) \le m-1$. Consequently $A_2(f) = \frac{1}{2}\sum_i \|Q_i\|_{\mathrm{F}}^2 \ge (T_f)_+^2/(2D_*)$ by Cauchy's inequality. This remains true if the actual test tensor vanishes.

For the regular reference,

$$T_* := \mathbb{E}V_*^2 - \beta^2 = \alpha^2\{\mathbb{E}[(\max_i Z_i^{\mathrm{iid}})^2] - 1\} > 0, \qquad A_2(g) = \frac{T_*^2}{2D_*}, \qquad T_*/D_* \le Cm.$$

For any $e \ge 0$, $(T_* - e)_+^2 \ge T_*^2 - 2T_*e$; hence (6.9)–(6.11) imply

$$A_2(f) - A_2(g) \ge -CL^3J - CL^2e^{-16L}.$$

The exterior floor (6.3) pays the absolute error: $L^2e^{-16L}/J \le Cm^{-13}L^{5/2} = o(1)$. Therefore

$$A_2(f) - A_2(g) \ge -CL^3J. \tag{6.12}$$

This is an exterior estimate for the localized assignment. It is not a global relative second-chaos inequality near the regular partition.

6.5 The entire signed higher tail

Define $\mathcal{H}_r(f) = \sum_{k\ge3}(-r)^k A_k(f)$, and let g_0 be the regular assignment associated with a closest regular centroid B_0. Its linear routing loss is

$$R_0 = w - \langle B_0, B\rangle = \frac{\Delta + d^2}{2} \ge 0.$$

Let $\varepsilon = \frac{1}{2}\|f - g_0\|_{L^1}$. On each point the total mass sent away from the regular winning label pays at least the top-two gap. Equation (2.11), followed by the elementary bathtub estimate, gives

$$\varepsilon \le \min\{1, C\sqrt{mR_0}\}. \tag{6.13}$$

This holds for fractional assignments too.

Put $\tau^2 = 1/(4L)$. Hypercontractivity and $p_i \le 3/(2m)$ give

$$\|T_\tau f\|_2^2 + \|T_\tau g_0\|_2^2 \le C/m.$$

For the difference, write $e_i = \mathbb{E}|f_i - g_{0,i}| \le C/m$. Since $|f_i - g_{0,i}| \le 1$,

$$\|T_\tau(f_i - g_{0,i})\|_2^2 \le e_i^{2/(1+\tau^2)} \le Ce_i/m.$$

Summation yields $\|T_\tau(f - g_0)\|_2^2 \le C\varepsilon/m$. On degrees at least three, factor the exact signed multiplier through T_τ; the residual operator has norm at most $(r/\tau^2)^3$. Cauchy's inequality therefore gives

$$|\mathcal{H}_r(f) - \mathcal{H}_r(g_0)| \le Cr^3L^3\frac{\sqrt{\varepsilon}}{m}. \tag{6.14}$$

All higher degrees remain in this estimate, with their signs; no finite chaos truncation is substituted.

If $J \ge d^2$, then $R_0 \le J$, and (6.13)–(6.14) and the exterior floor give

$$\frac{|\mathcal{H}_r(f) - \mathcal{H}_r(g_0)|}{J} \le \frac{Cr^3L^3}{m^{3/4}J^{3/4}} \le Cr^3m^{3/2}L^{27/8}.$$

If $J < d^2$, then $R_0 \le d^2$. Use $d = \alpha q$, $J \ge cq/m^3$, and $q \ge c/\sqrt{L}$ to obtain

$$\frac{|\mathcal{H}_r(f) - \mathcal{H}_r(g_0)|}{J} \le Cr^3m^{7/4}L^{13/4}q^{-1/2} \le Cr^3m^{7/4}L^{7/2}.$$

The latter also dominates the first case. Rotation invariance of the regular Hermite weights gives

$$|\mathcal{H}_r(f) - \mathcal{H}_r(g)| \le Cr^3m^{7/4}L^{7/2}J. \tag{6.15}$$

6.6 Completion of the proof

Hermite orthogonality gives exactly

$$S_{-r}(f) - S_{-r}(g) = D + r^2\{A_2(f) - A_2(g)\} + \mathcal{H}_r(f) - \mathcal{H}_r(g).$$

Equations (6.12) and (6.15), with $J = D/r$, therefore imply

$$S_{-r}(f) - S_{-r}(g) \ge D\{1 - CrL^3 - Cr^2m^{7/4}L^{7/2}\} \ge D/2 > 0 \tag{6.16}$$

for sufficiently large m, uniformly in $0 < r \le 1/(m-1)$. The two endpoint errors are $O(L^3/m)$ and $O(L^{7/2}/m^{1/4})$. Strictness follows from $D = rJ > 0$ in the exterior. This contradicts competitiveness. Consequently every competitive assignment lies inside $d \le 1/m$, where Prior input 2.5 forces the regular assignment. Noncompetitive assignments already have strictly larger noise value. This proves both the inequality and its equality classification in Theorem 1.1, directly for all measurable simplex-valued assignments. No full-noise attainment, purity, stationarity, interface regularity, or limiting optimization argument is required. All thresholds are independent of finite N and of positive r in the voting interval. □

7 Discrete plurality and the fixed-alphabet cut threshold

This section applies the primary transfer results of Isaksson–Mossel [19], with the precise negative-range and single-rule bookkeeping in the audited companion [17]. Its conclusions inherit the proof-candidate status of Theorem 1.1. Write $s_m(\rho) = S_\rho(g)$.

On $[m]^k$ use uniform product measure. The coordinate noise channel is

$$\mathbb{P}\{Y_j = a \mid X_j = b\} = \rho\mathbf{1}_{\{a=b\}} + (1-\rho)/m, \tag{7.1}$$

independently for $j \le k$. Its allowed range is $-1/(m-1) \le \rho \le 1$. The stability of a simplex-valued rule is $\mathbb{E}\langle F(X), F(Y)\rangle$. With any orthonormal product Fourier basis, put

$$\mathrm{Inf}_j^{\le d_0}(F_i) = \sum_{\substack{\sigma_j \ne 0 \\ |\operatorname{supp}\sigma| \le d_0}} \widehat{F}_i(\sigma)^2.$$

The ordinary influence is the same sum without the degree restriction.

Corollary 7.1 (Negative Plurality is Stablest)**.** *For every fixed $m \geq m_0$, every $\rho \in [-1/(m-1), 0)$ and every $\varepsilon > 0$, there are $d_0 \geq 1$ and $\tau > 0$, depending only on (m, ρ, ε), with the following property. For every voter count $k \geq 1$ and every $F : [m]^k \to \Delta_m$ satisfying $\mathrm{Inf}_j^{\leq d_0}(F_i) \leq \tau$ for every i, j,*

$$S_\rho(F) \geq s_m(\rho) - \varepsilon = \lim_{\ell \to \infty} S_\rho(\mathrm{Plurality}_{\ell,m}) - \varepsilon.$$

No output-balance, label-equivariance, unanimity, or purity assumption is imposed. The influence parameters are independent of k; no explicit or uniform-in-m influence threshold is claimed.

Proof. Theorem 7.1 of [19], at the fixed parameters in the statement, transfers F to a pure Gaussian assignment h in dimension $k(m-1)$ with the same masses and $|S_\rho(h) - S_\rho(F)| \leq \varepsilon$. This finite dimension is at least $m-1$, so Theorem 1.1 applies. Its absence of a mass constraint is essential here. Lemma 7.3 of [19] identifies the Gaussian regular value with the plurality limit by the joint central limit theorem; Gaussian ties have measure zero, so the finite plurality tie convention does not affect the limit. These two statements give the displayed inequality. □

The stronger ordinary-influence assumption also suffices. So does a bound by τ on the sum of the ordinary influences of all output labels at each coordinate. The negative endpoint of (7.1) is included: a coordinate then changes uniformly to a different label. At $\rho = 0$, the conclusion follows without any influence restriction from $S_0(F) = \|\mathbb{E}F\|^2 \geq 1/m$. There is no implication here for positive noise or for a channel below the negative endpoint.

7.1 The exact Frieze–Jerrum constant

Define

$$\mathfrak{a}_m = \min_{-1/(m-1) \leq \rho \leq 0} \frac{m}{m-1} \frac{1 - s_m(\rho)}{1 - \rho}. \tag{7.2}$$

This is an exact fixed-m constant, and its minimum exists by continuity. We do not assert that a minimizing correlation is the left endpoint. The value of the ratio at zero is 1.

For $0 \leq \rho < 1$, purity of g and its nonnegative Hermite weights give

$$s_m(\rho) = \frac{1}{m} + \sum_{j \geq 1} \rho^j A_j(g) \leq \frac{1}{m} + \rho\left(1 - \frac{1}{m}\right).$$

Thus the regular ratio is at least 1 throughout the positive interval. Its usual Frieze–Jerrum infimum over the full channel interval is therefore exactly (7.2). This is the negative-range reduction of [19, Lemma A.4]; no positive-noise simplex optimality is needed.

Corollary 7.2 (Sharp fixed-m MAX-m-CUT threshold under UGC)**.** *For every fixed integer $m \geq m_0$ and every $\varepsilon > 0$, there is a polynomial-time algorithm for nonnegative rational edge-weighted MAX-m-CUT with approximation ratio $\mathfrak{a}_m - \varepsilon$. Assuming the Unique Games Conjecture, approximating this problem within $\mathfrak{a}_m + \varepsilon$ is NP-hard. The algorithm and reduction may depend on the fixed m, ε.*

Proof. An edge of the Frieze–Jerrum semidefinite relaxation has vector correlation $\rho \geq -1/(m-1)$ and SDP contribution $(m-1)(1-\rho)/m$. Regular-simplex rounding cuts it with probability $1 - s_m(\rho)$. Equation (7.2) and the positive-range estimate above give the edgewise ratio $\mathfrak{a}_m$. Polynomial-precision solution and rounding give arbitrary small slack, as in [19, Appendix A.3, Theorem A.6].

For hardness, the UGC reduction in [19, Appendix A.4], in its negative-correlation form, has completeness arbitrarily close to $(m-1)(1-\rho)/m$ and soundness arbitrarily close to $1-\Lambda_m^-(\rho)$, where Λ_m^- is the unrestricted Gaussian minimum. Theorem 1.1 identifies it with $s_m(\rho)$ for every negative channel correlation. Dimensions below $m-1$ are covered by cylindrical embedding. Choose a correlation at, or sufficiently close to, a minimizer of (7.2), then choose the reduction's errors sufficiently small. Rational correlations suffice by continuity. The resulting hardness ratio is arbitrarily close to $\mathfrak{a}_m$ from above.

The soundness calculation is genuinely single-rule. Conditional on a verifier vertex v, its independently sampled neighbors define the averaged simplex-valued function $G_v(x) = \mathbb{E}_w[\mathbf{1}_{\{F_w(\sigma_{vw}x)=i\}}]_{i=1}^m$. The rejection probability is exactly $\mathbb{E}_v S_\rho(G_v)$. Corollary 7.1 supplies the low-degree soundness and decoding step for this same function in both slots. No inequality for two independently chosen Gaussian rules is used. □

The approximation statement is unconditional; UGC is needed only for the hardness assertion. The same sharp threshold for simple unweighted graphs follows from the fixed-m mixing-graph reduction proved in [17, Section 4], using the construction in [22, Theorem 13]; the direct statement above uses the primary weighted formulation. Neither conclusion is a running-time statement uniform in an input alphabet size tending to infinity. The older asymptotic endpoint result [20, Proposition 11.4] does not by itself identify the exact constant (7.2) for every fixed sufficiently large m.

A Frozen sources and proof provenance

The authoritative internal theorem statement is [14], with the direct competitive proof [16]. The endpoint and remainder equality facts are proved explicitly from covariance concavity in Proposition 2.2, following [21]; the older endpoint references in Section 10 of [1] are not used. The new proofs in this manuscript correspond to [11, 12, 13]; the radial premise left conditional in the last of these is proved here in Theorem 4.1.

The entries below identify the exact bytes used. Internal review is not a substitute for independent mathematical review of these sources and their dependencies. Copies of the internal frozen sources and a machine-readable hash manifest accompany the editable manuscript. Copyright primary articles are not bundled; their external URLs and private-cache hashes are recorded instead. No attribution of human authorship is made for internal source files.

The unchanged companion `dependency_review/` supplies the section-level transitive proof closure: 34 internal sources, their exact used sections, and an external-input ledger. Its report is a provenance and completeness check against the frozen first-edition snapshot, not a new external correctness audit. This edition incorporates its endpoint replacement and external-reference corrections explicitly.

`dependency_review/SOURCE_MAP.json`

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